TED(21) – 5023A **REVISION 2021**

MODEL QUESTION PAPER

Modern Production Processes

Time: 3 Hours **Max Marks: 75**

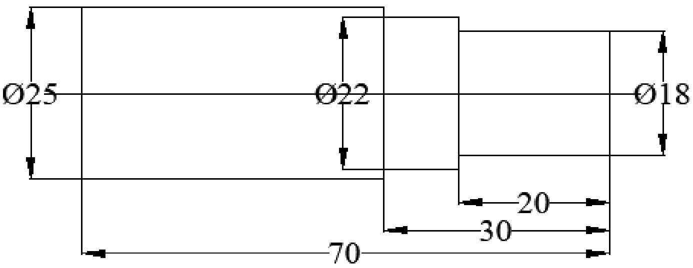
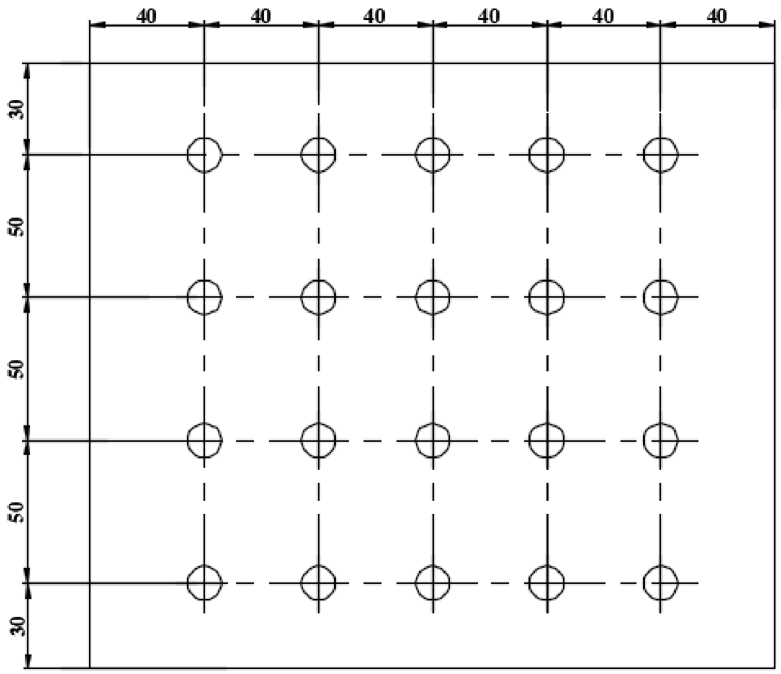
I. Answer all the following questions **(9 x 1 = 9 Marks)**

1	Selecta type of jig suitable for drilling a series of equidistant holes on cylindrical components.	M 1.01	A
2	The process of coating of thin film deposition where a solid material is vaporized in a vacuum environment and deposited on substrates is called…………………	M 1.03	R
3	Name the process of the metal is removed by the controlled dissolution of the anode according to Faraday's law of electrolysis.	M 2.01	R
4	Choose a non-conventional machining method for making complex die profiles.	M 2.02	A
5	In AJM set-up the distance from work surface to tip of the nozzle is called ………	M 2.02	R
6	CNC Lathes are programmed …………. number of axes.	M 3.01	R
7	Write the code will change specified input values in millimetres.	M 3.02	R
8	Select the manufacturing approach of using computers to control entire production process.	M 4.01	A
9	Write two applications of industrial robots	M 4.03	A

II.Answer any Eight questions from the following **(8 x 3= 24 Marks)**

1	List important design considerations in design of jigs and fixtures	M 1.01	R
2	Describe the principle of Chemical Vapour deposition	M 1.03	R
3	What are the advantages of Metal Spraying?	M1.03	R
4	Write two applications of Non-conventional machining processes of the following: a) EDM b) USM c) LBM	M2.03	A
5	Draw schematic arrangement of AWJM and label its components.	M 2.02	R
6	Write short note on Computer Aided Process Planning.	M 4.01	U
7	Describe Direct Numerical Control	M 4.01	U
8	List the benefits of group technology	M4.02	U
9	Discuss Automated Guided Vehicle System in FMS	M4.02	U
10	List the applications of industrial robots	M 4.04	A

III. **Answer all questions from the following** **(6x 7 = 42 Marks)**

1	Illustrate the fixtures used for turning, drilling and milling	M 1.02	U
	OR		
2	Explain the different stages of manufacturing of powder metallurgy component	M 1.04	U
3	Enumerate the classification of Unconventional machining process	M 2.01	R
	OR		
4	Explain the process of LBM with a neat sketch.	M 2.02	U
5	Explain the working principle, elements and characteristics of wire EDM with neat sketch.	M2.02	U
	OR		
6	Discuss the advantages and limitations of non- conventional machining process over conventional machining processes.	M2.03	U
7	Prepare a CNC part programming from a 25 mm diameter mild steel rod to turn machine component as shown in figure. Explain the important functions used in the CNC code. 	M 2.04	A
	OR		
8	Write a CNC part program to drill 20 holes of diameter of 12mm each in a machine component as shown in the figure. The raw material to be employed is mild steel plate of 6 mm thickness. Explain the important functions used in the CNC code. 	M 2.07	A

9	Explain the classification of CNC machines	M 3.01	U
	OR		
10	Describe various steps involved in rapid prototyping process chain	M 3.03	U
11	Explain any three layouts of Flexible Manufacturing System with diagrams	M 4.02	R
	OR		
12	Explain the four common Robot configurations with neat sketch	M 4.03	R